Exp. No: 14	Analysis of EDFA-Based WDM Link Performance using OptiSystem

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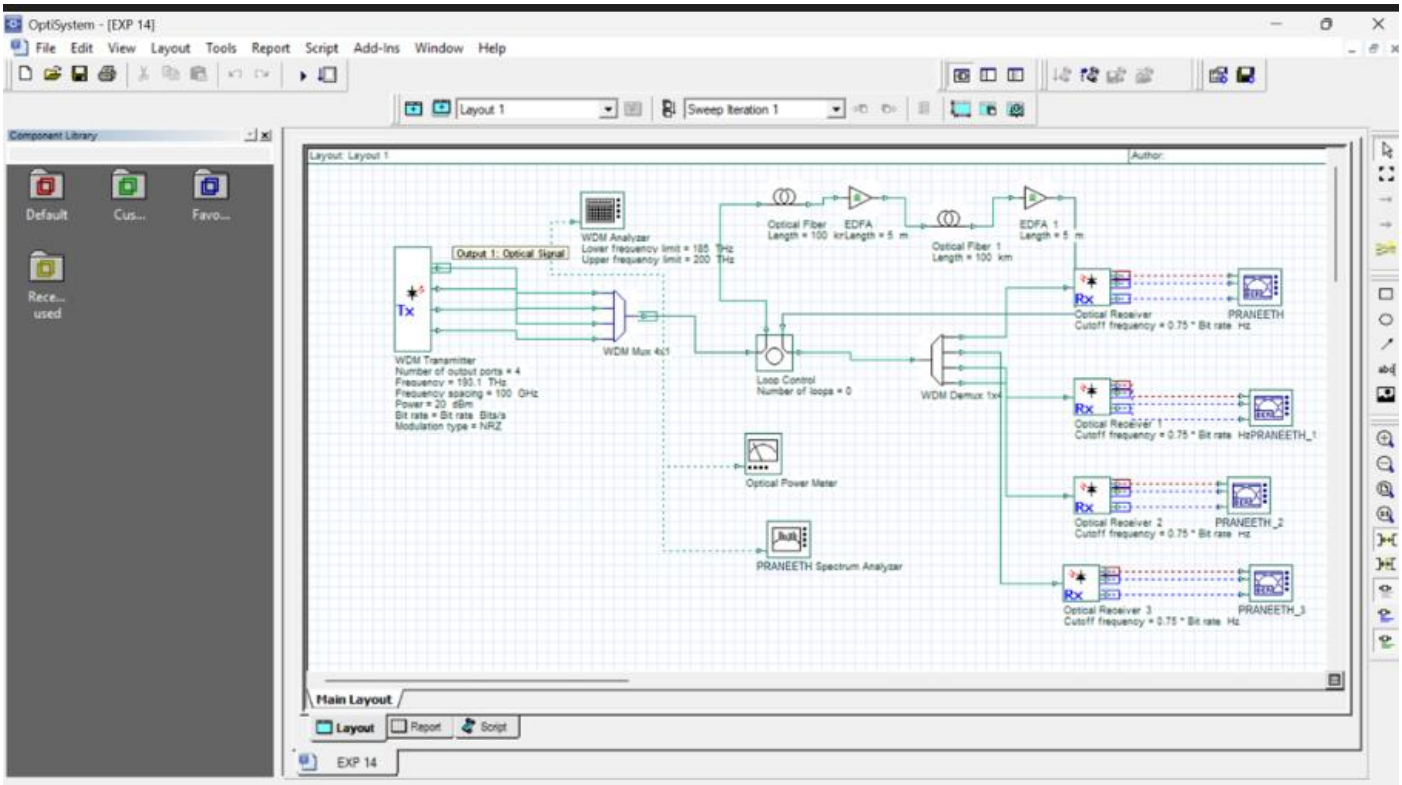
OBJECTIVES

1. To design and simulate a Dense Wavelength Division Multiplexing (DWDM) optical communication system using OptiSystem.
2. To investigate the impact of varying optical fiber length on overall system performance.
3. To examine the effectiveness of an Erbium-Doped Fiber Amplifier (EDFA) in compensating for fiber attenuation losses.
4. To evaluate and compare received power, bit error rate (BER), and Q-factor across different fiber lengths.

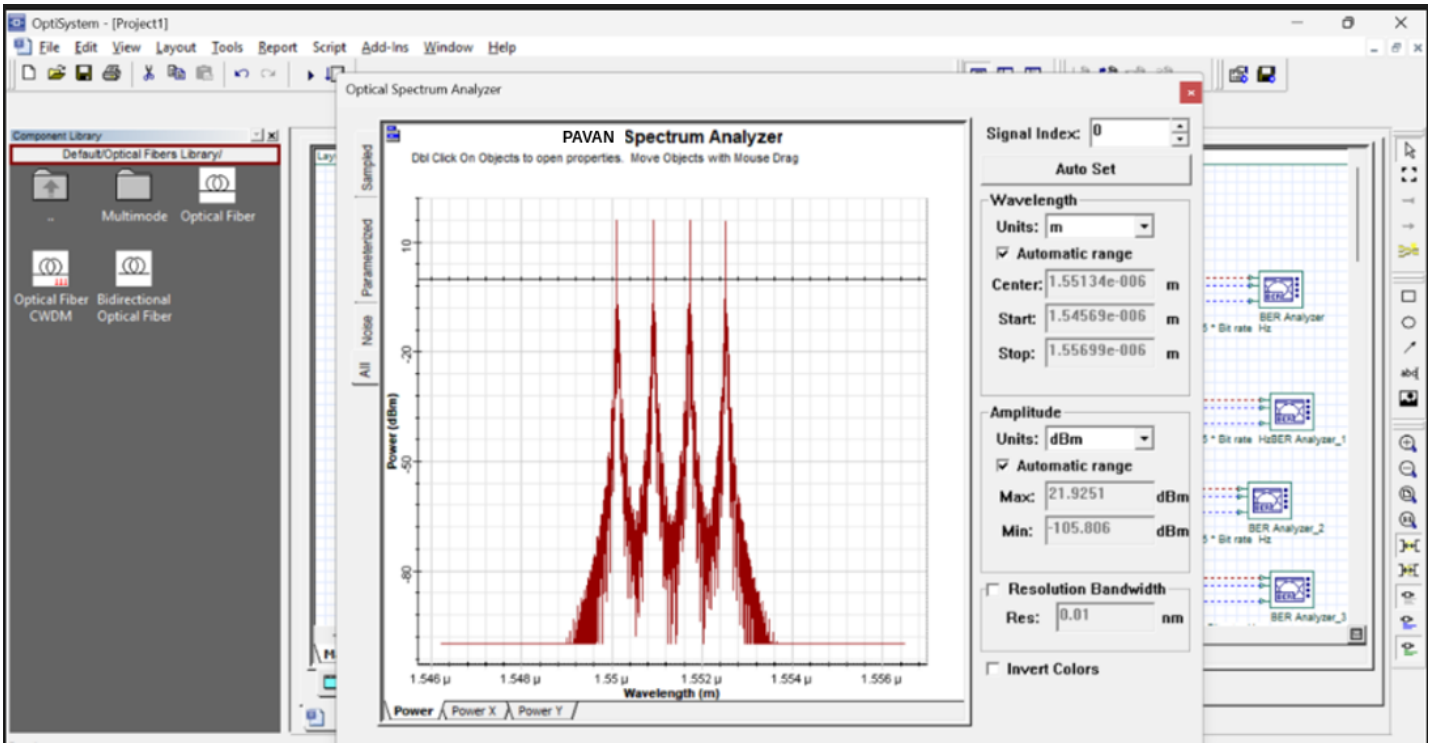
SOFTWARE REQUIRED

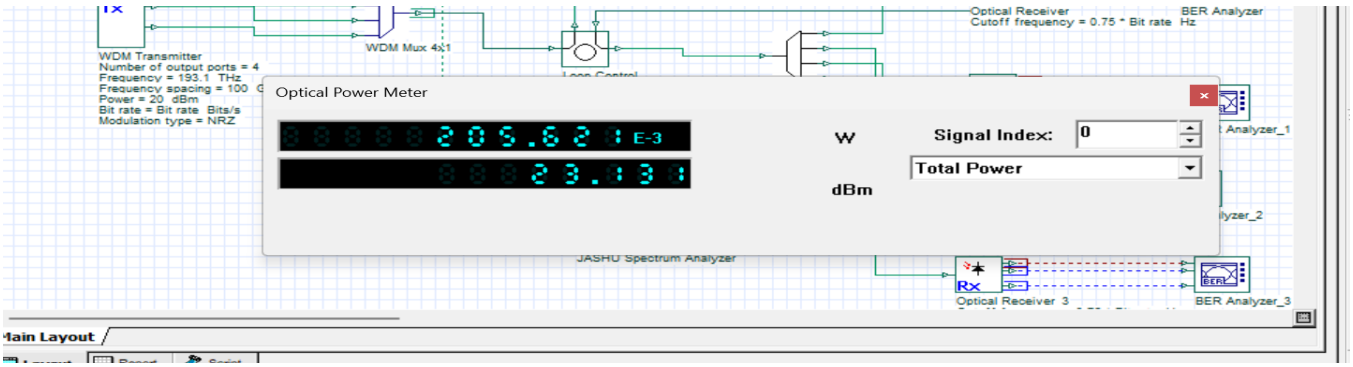
- Optisystem software

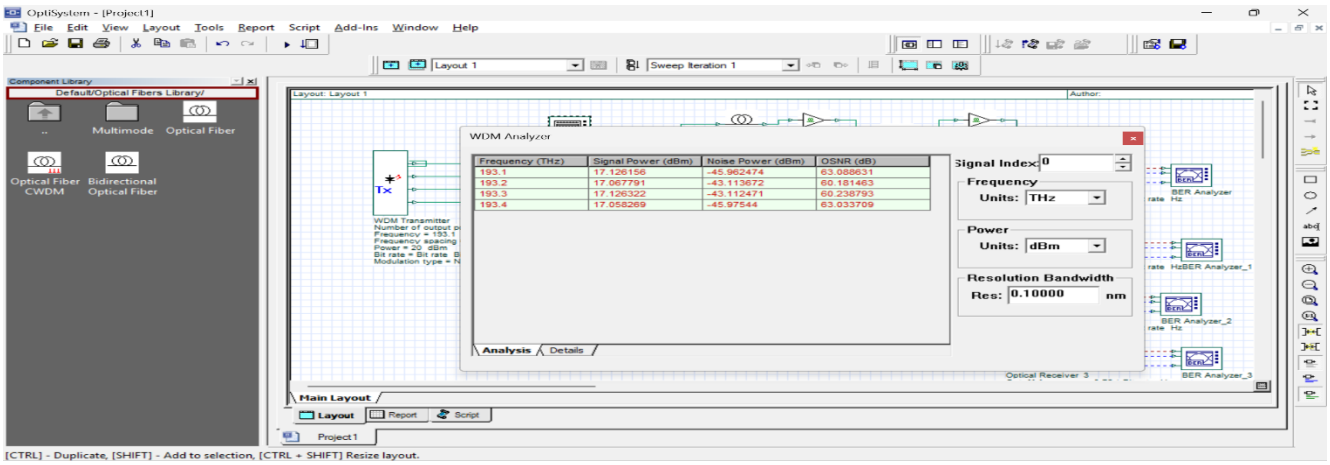
BLOCK DIAGRAM

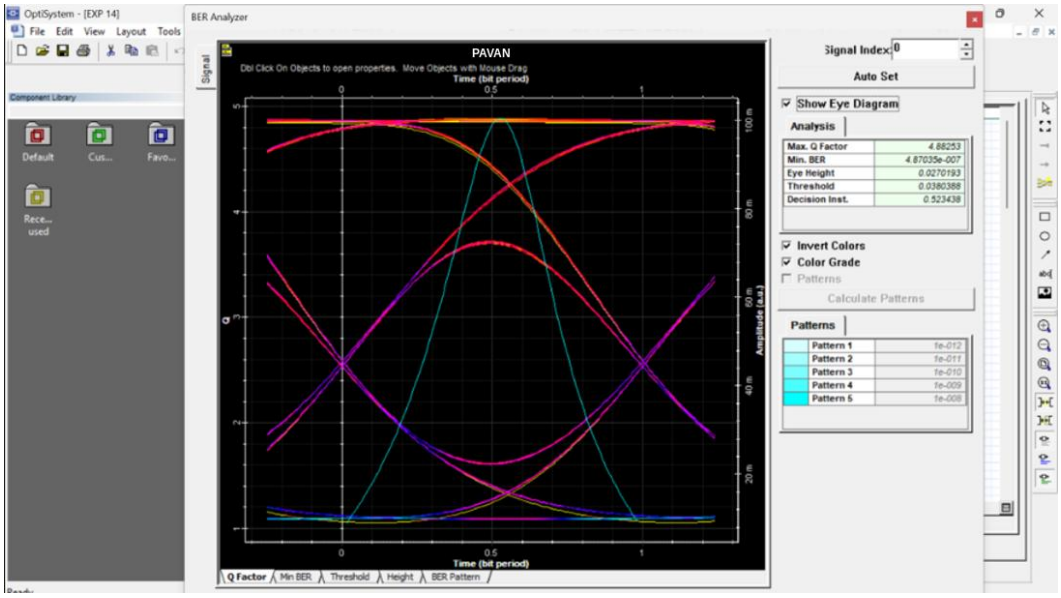


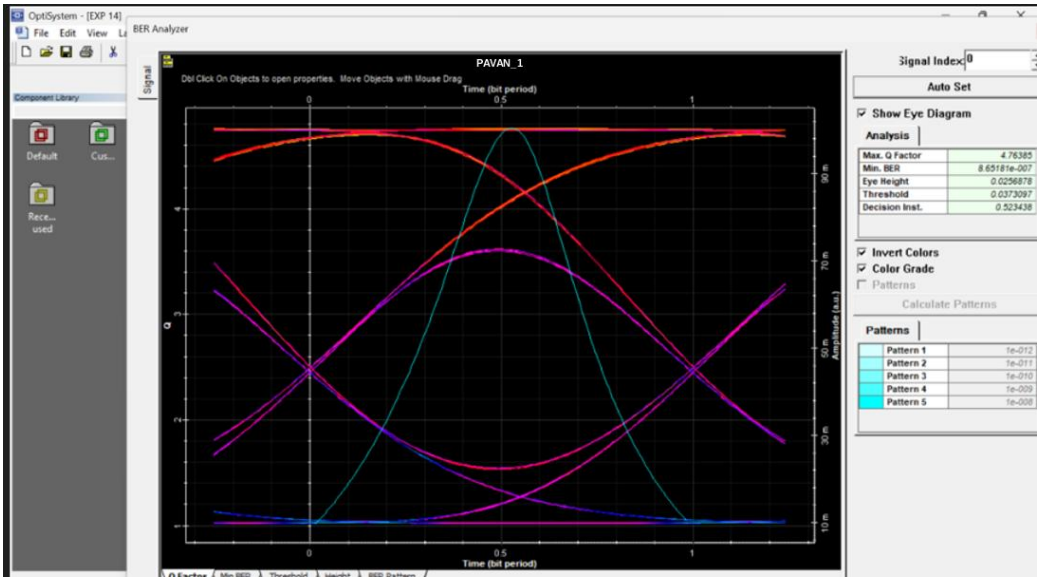
OUTPUTS

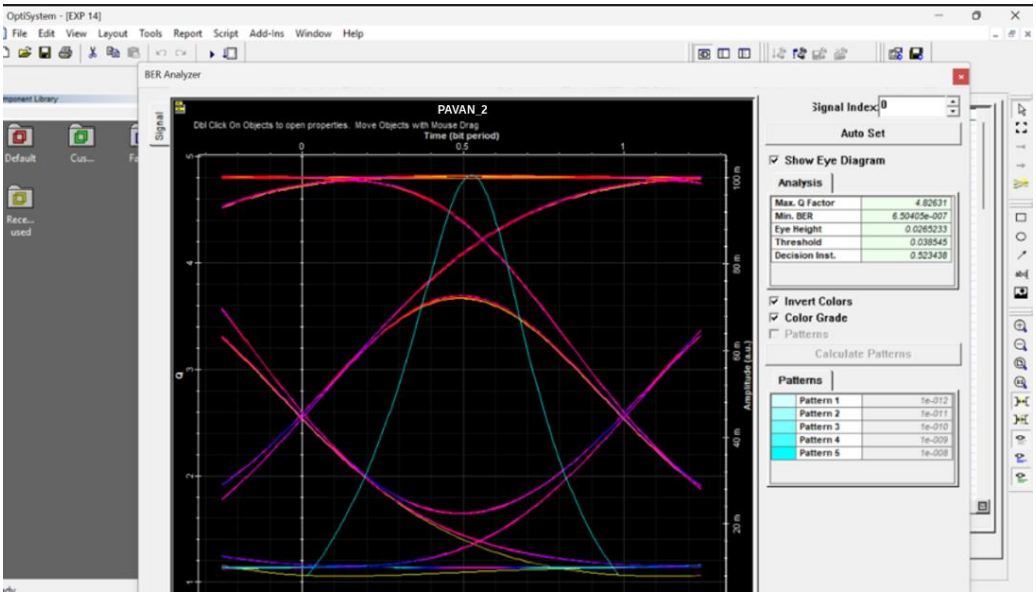

OptiSystem - [Project1]
File Edit View Layout Tools Report Script Add-Ins Window Help
Optical Spectrum Analyzer
PAVAN Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55134e-006 m
Start: 1.54569e-006 m
Stop: 1.55699e-006 m
Amplitude
Units: dBm
Automatic range
Max: 21.9251 dBm
Min: -105.806 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors
Power (dBm)
Wavelength (m)
Power
Power X
Power Y
Component Library
Default/Optical Fibers Library/
Multimode
Optical Fiber
Optical Fiber CWDM
Bidirectional Optical Fiber
BER Analyzer

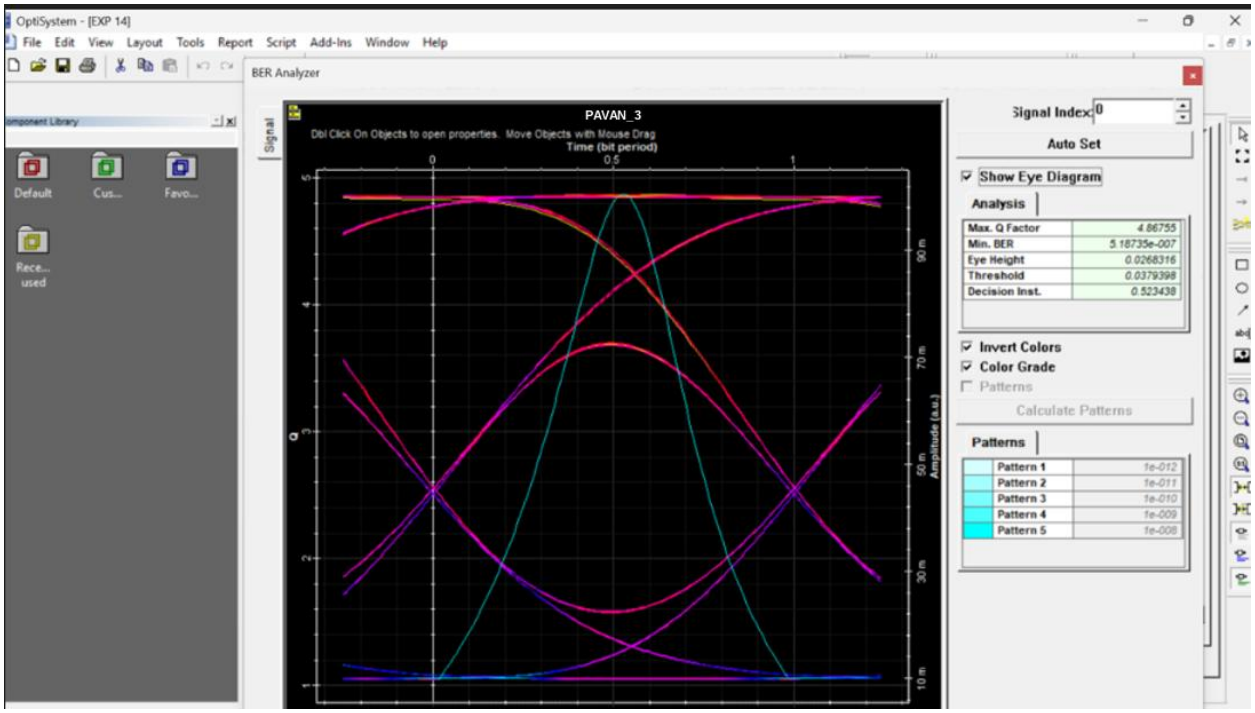

WDM Transmitter
Number of output ports = 4
Frequency = 193.1 THz
Frequency spacing = 100
Power = 20 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
WDM Mux 4x1
Optical Receiver
Cutoff frequency = 0.75 * Bit rate
BER Analyzer
Hz
Optical Power Meter
205.621 E-3
23.131
W
dBm
Signal Index: 0
Total Power
JASHU Spectrum Analyzer
Optical Receiver 3
BER Analyzer_3
Main Layout


OptiSystem - [Project1]
File Edit View Layout Tools Report Script Add-Ins Window Help
Component Library
Default/Optical Fibers Library/
Multimode
Optical Fiber
Optical Fiber CWDM
Bidirectional Optical Fiber
WDM Analyzer
Frequency (THz) Signal Power (dBm) Noise Power (dBm) OSNR (dB)
193.1 17.126156 -45.962474 63.088631
193.2 17.067791 -43.113672 60.181463
193.3 17.126322 -43.112471 60.238793
193.4 17.058269 -45.97544 63.033709
Signal Index: 0
Frequency
Units: THz
Power
Units: dBm
Resolution Bandwidth
Res: 0.10000 nm
Analysis
Details
Main Layout
Layout
Report
Script
Project1
[CTRL] - Duplicate, [SHIFT] - Add to selection, [CTRL + SHIFT] Resize layout.


OptiSystem - [EXP 14]
File Edit View Layout Tools
BER Analyzer
PAVAN
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.88253
Min. BER 4.87035e-007
Eye Height 0.0270193
Threshold 0.0380388
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


OptiSystem - [EXP 14]
BER Analyzer
PAVAN_1
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.76385
Min. BER 8.65181e-007
Eye Height 0.0256878
Threshold 0.0373097
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008


OptiSystem - [EXP 14]
File Edit View Layout Tools Report Script Add-Ins Window Help
BER Analyzer
PAVAN_2
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal Index 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.82631
Min. BER 6.50405e-007
Eye Height 0.0265233
Threshold 0.038545
Decision Inst. 0.523438
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008



Reflection

I learned how an EDFA amplifies optical signals in a WDM communication system. I understood how optical amplifiers compensate for transmission losses and improve the performance of optical communication links.

CONCLUSION:

The DWDM system with EDFA was analyzed for varying fiber lengths. Longer fibers increase attenuation and degrade the signal, while the EDFA compensates for this loss, improving received power, Q-factor, and BER. This confirms DWDM with optical amplification is suitable for high-capacity, long-distance communication.

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightag e
System Design & DWDM Setup	Correct selection and connection of OptiSystem components for the DWDM link (PRBS Generator, NRZ Pulse Generator, CW Lasers, WDM Multiplexer, Optical Fiber, EDFA, WDM Demultiplexer, PIN Photodetectors, Low Pass Filters, BER Analyzers).	All four transmitter channels, multiplexer, fiber, EDFA, demultiplexer and receiver chains are correctly selected, connected in sequence and configured without errors.	All components are present with minor connection or layout errors, but the DWDM link remains functional.	Component selection is partially complete; missing receiver chains or EDFA placement errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuration & Execution	Accurate setup of input parameters: Transmit Power (20 dBm), Center Frequency (193.1 THz), Channel Spacing (100 GHz), Fiber Length (50 km, 75 km, 100 km) and Loop Control (0).	All input parameters and fiber-length test cases are accurately set and executed, with and without the EDFA.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but multiple fiber-length cases or the with/without EDFA comparison are missing.	Incorrect parameter settings or failure to run the required test cases.	20%
Measurement & Performance Monitoring	Accurate extraction and recording of Min BER, Max Q-Factor, received power, optical spectrum and eye diagrams for all channels and fiber lengths.	Precise data extraction across all fiber lengths, neatly tabulated with complete BER, Q-factor and power readings and clear spectrum/eye diagrams.	Data is collected with minor omissions in BER, Q-factor or power readings.	Partial data collection with incomplete tables or missing spectrum/eye diagram captures.	Inaccurate, incomplete or missing data and measurements.	20%
Amplification & Fiber Loss Analysis	Detailed evaluation of fiber attenuation, dispersion, EDFA gain and noise, and the effect of amplification on received power, Q-factor and BER.	Comprehensive analysis correctly relating fiber length, attenuation and EDFA compensation to received power, Q-factor and BER, with and without amplification.	Clear analysis identifying basic trends in power, BER and Q-factor relative to fiber length and EDFA use.	Basic attempt at result analysis without explaining physical mechanisms such as attenuation, dispersion or amplifier gain.	Erroneous or missing analysis of simulation results.	20%
Report & Conclusion	Identification of the benefit of EDFA amplification, achievable transmission distance, trade-offs (amplifier noise, channel spacing), structure and clarity of the report.	Report features clear spectra and eye diagrams, complete result tables and insightful conclusions identifying optimal operating parameters for the DWDM link.	Satisfactory report with relevant displays and a sound concluding summary.	Incomplete report with missing graphs, readings or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%